

2009 HCI Preliminary Examinations H2 Mathematics Paper 1 Solutions

1.

Let P_n be the proposition that $u_n = \frac{a(n+1)n}{(n-1)!}$ for $n \in \mathbb{N}^+$.

For $n = 1$, LHS of $P_1 = 2a$

$$\text{RHS of } P_1 = \frac{a(1+1)1}{(1-1)!} = \frac{2a}{1} = 2a$$

$$\text{LHS of } P_1 = \text{RHS of } P_1$$

Thus, P_1 is true.

Assume that P_k is true for some $k \in \mathbb{N}^+$.

$$\text{i.e. } u_k = \frac{a(k+1)k}{(k-1)!}$$

To prove that P_{k+1} is true:

$$\begin{aligned} u_{k+1} &= \frac{k+2}{k^2} u_k \\ &= \frac{(k+2)}{k^2} \frac{a(k+1)k}{(k-1)!} \\ &= \frac{a(k+2)(k+1)}{k(k-1)!} \\ &= \frac{a(k+2)(k+1)}{k!} \end{aligned}$$

Thus P_{k+1} is true.

Since P_1 is true, and P_k true $\Rightarrow P_{k+1}$ is true. By Mathematical

Induction, P_n is true for all $n \in \mathbb{N}^+$.

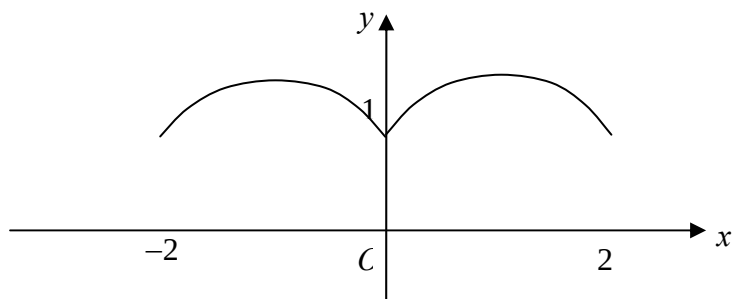
Limit of sequence is 0.

2.

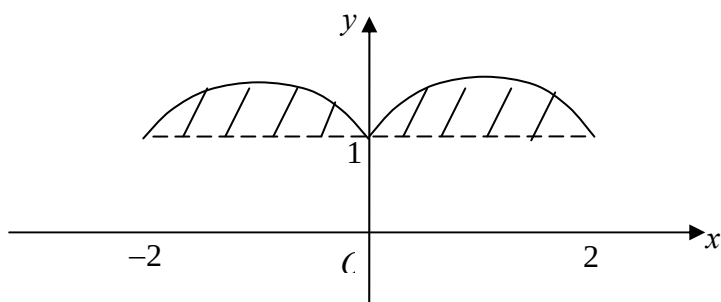
Given curve C is defined by:

$$x = 2 \sin t - \sin 2t, \quad y = 2 \cos t - \cos 2t, \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

(i) The graph of C is



(ii)



Area of the shaded region

$$= \int_{-2}^2 y dx - 4 \times 1$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (2 \cos t - \cos 2t)(2 \cos t - 2 \cos 2t) dt - 4$$

$$= 1.42478$$

$$= 1.4 \text{ (correct to 1 d.p.)}$$

3(i)

Concave downwards $\Rightarrow f''(x) < 0$ so the gradient of $f'(x) < 0$

Strictly increasing $\Rightarrow f'(x) > 0$

$x > 1$ (ans)

(ii)

Stationary points: $f'(x) = 0 \Rightarrow x = -3, 0$

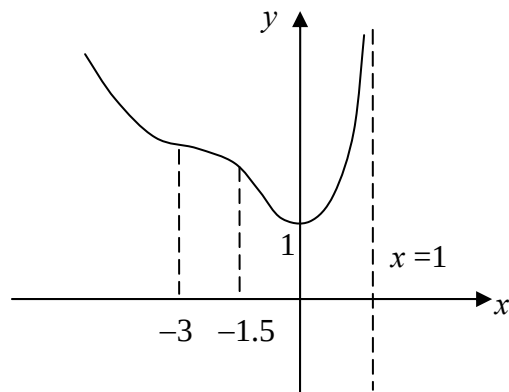
x	$(-3)^-$	-3	$(-3)^+$
$f'(x)$	-ve	0	-ve
	$\searrow$	---	$\searrow$

Point of inflexion
at $x = -3$

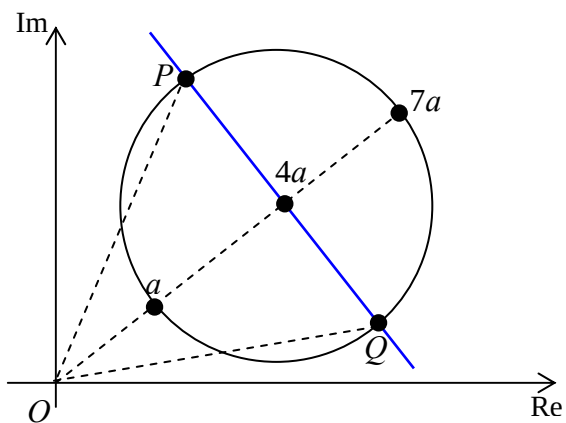
x	0^-	0	0^+
$f'(x)$	-ve	0	+ve
	$\searrow$	---	$\nearrow$

Minimum point at
 $x = 0$.

(iii)



4



If $\arg\left(\frac{p}{q}\right) > 0$,

$$\Rightarrow \arg(p) - \arg(q) > 0 \Rightarrow \arg(p) > \arg(q)$$

$$\arg\left(\frac{p}{q}\right) = \angle POQ = 2\alpha = 2 \tan^{-1} \frac{3|a|}{4|a|} = 2 \tan^{-1} \frac{3}{4}$$

If $\arg\left(\frac{p}{q}\right) < 0$, then $\arg\left(\frac{p}{q}\right) = -2 \tan^{-1} \frac{3}{4}$.

$$\begin{aligned} |p+q| &= 2 \times (\text{distance between } O \text{ and centre of the circle}) \\ &= 2|4a| = 8|a| \end{aligned}$$

5

$$g(r) = \frac{1}{(3r-2)(3r+1)(3r+4)} = \frac{a}{(3r-2)(3r+1)} - \frac{b}{(3r+1)(3r+4)}$$

$$1 = a(3r+4) - b(3r-2)$$

$$\text{Solving, we get } a = b = \frac{1}{6}$$

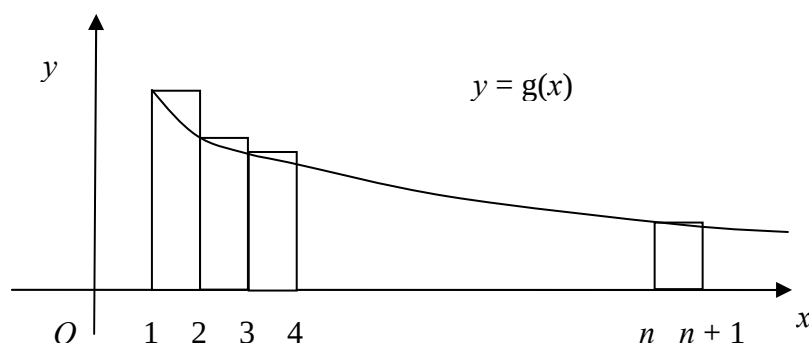
$$(i) \sum_{r=1}^n g(r) = \frac{1}{6} \left(\begin{array}{c} f(1) - f(2) \\ + f(2) - f(3) \\ \dots\dots\dots \\ + f(n) - f(n+1) \end{array} \right) = \frac{1}{6} (f(1) - f(n+1))$$

$$\sum_{r=1}^n g(r) = \frac{1}{6} \left(\frac{1}{4} - \frac{1}{(3n+1)(3n+4)} \right)$$

$$(ii) \frac{1}{(3n-5)(3n-2)(3n+1)} + \frac{1}{(3n-2)(3n+1)(3n+4)} + \dots$$

$$= \sum_{r=n-1}^{\infty} g(r) = \left(\sum_{r=1}^{\infty} g(r) - \sum_{r=1}^{n-2} g(r) \right)$$

$$= \frac{1}{6} \left(\frac{1}{4} \right) - \frac{1}{6} \left(\frac{1}{4} - \frac{1}{(3n-5)(3n-2)} \right) = \frac{1}{6(3n-5)(3n-2)}$$



$\sum_{r=1}^n g(r)$ represents the sum of area of rectangles drawn with breadth

of unit 1 and height $g(r)$ for $r = 1, 2, 3, \dots, n$ as shown in diagram

while $\int_1^{n+1} g(x) dx$ represents the area under the curve bounded by

$r = 1$ to $r = n + 1$ which is smaller than the sum of area of rectangles.

6

(i) $(ay)^2 = x(4-x) \dots (1)$

$$2a(ay) \frac{dy}{dx} = 4-2x \Rightarrow \frac{dy}{dx} = \frac{4-2x}{2a^2y} = \frac{2-x}{a^2y}$$

For tangents // to the y-axis, $\frac{dy}{dx}$ is undefined.

$$\Rightarrow y = 0$$

Sub. $y = 0$ in (1), $0 = x(4-x)$

$$\Rightarrow x = 0, x = 4$$

(ii) $A = \text{Area of } PQRS = 2y(PQ)$

$$y = 2\sqrt{x(4-x)}, PQ = 4-2x$$

(symmetrical abt $x = 2$)

$$A = 8(2-x)\sqrt{4x-x^2}$$

$$\begin{aligned} \frac{dA}{dx} &= 8 \left[-\sqrt{4x-x^2} + \frac{1}{2}(2-x) \frac{4-2x}{\sqrt{4x-x^2}} \right] \\ &= \frac{8}{\sqrt{4x-x^2}} \left[-(4x-x^2) + (2-x)^2 \right] \end{aligned}$$

For $\frac{dA}{dx} = 0$, $-(4x-x^2) + (2-x)^2 = 0$

$$-4x + x^2 + 4 - 4x + x^2 = 0$$

$$2x^2 - 8x + 4 = 0 \Rightarrow x^2 - 4x + 2 = 0$$

$$x = \frac{4 \pm \sqrt{8}}{2} = 2 \pm \sqrt{2}$$

Since $x < 2$, $x = 2 - \sqrt{2}$.

Check maximum using first derivative test.

x	$(2-\sqrt{2})^-$	$2-\sqrt{2}$	$(2-\sqrt{2})^+$
$f'(x)$	+ve	0	-ve
			

A is maximum when $x = 2 - \sqrt{2}$.

7.

$$(i) f(x) = \frac{3x^2 - 3x + 4}{(2x-1)(x^2+3)}$$

$$\text{Let } \frac{3x^2 - 3x + 4}{(2x-1)(x^2+3)} = \frac{A}{2x-1} + \frac{Bx+C}{x^2+3}$$

$$\Rightarrow 3x^2 - 3x + 4 = A(x^2 + 3) + (Bx + C)(2x - 1)$$

$$\text{Put } x = \frac{1}{2} \Rightarrow A = 1$$

$$\text{Equate coefficients of } x^2 : B = 1$$

$$\text{Equate constant terms : } C = -1$$

$$\therefore \frac{3x^2 - 3x + 4}{(2x-1)(x^2+3)} = \frac{1}{2x-1} + \frac{x-1}{x^2+3} \quad (\text{Ans})$$

$$= \left[-(1-2x) \right]^{-1} + (x-1) \left[3 \left(1 + \frac{x^2}{3} \right) \right]^{-1}$$

$$= - \left[1 + (-1)(-2x) + \frac{(-1)(-2)}{2!} (-2x)^2 + L \right]$$

$$+ \frac{1}{3} (x-1) \left[1 + (-1) \left(\frac{x^2}{3} \right) + L \right]$$

$$= -1 - 2x - 4x^2 - L - \frac{1}{3} + \frac{1}{3}x + \frac{x^2}{9} + L$$

$$= -\frac{4}{3} - \frac{5}{3}x - \frac{35x^2}{9} + L \quad (\text{Ans})$$

The expansion is valid when

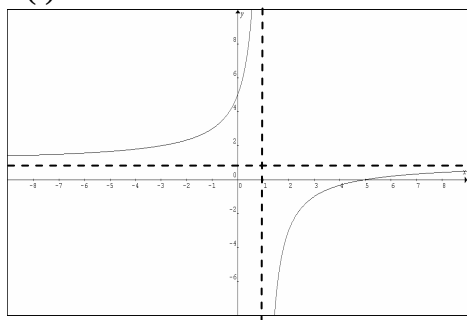
$$|2x| < 1 \quad \text{and} \quad \left| \frac{x^2}{3} \right| < 1 \quad \text{---[1]}$$

$$\text{i.e. } |x| < \frac{1}{2} \quad \text{and} \quad |x| < \sqrt{3}$$

$$\text{i.e. } -\frac{1}{2} < x < \frac{1}{2} \quad \text{---[2]} \quad (\text{Ans})$$

$$\begin{aligned} \text{Coefficient of } x^{2n} &= -2^{2n} - \frac{1}{3} \left(-\frac{1}{3} \right)^n \\ &= -2^{2n} + \left(-\frac{1}{3} \right)^{n+1} \quad (\text{Ans}) \end{aligned}$$

8(i)



$$f(x) = \frac{5-x}{1-x} = 1 + \frac{4}{1-x}$$

Since any horizontal line, $y = k$ cuts the graph of f at at most 1 point,

f is 1-1. $\therefore f^{-1}$ exists.

$$\text{Let } y = 1 + \frac{4}{1-x} \Rightarrow x = 1 - \frac{4}{y-1}.$$

$$f^{-1}(x) = 1 - \frac{4}{x-1} = 1 + \frac{4}{1-x}, x \in \mathbb{R} \setminus \{1\}$$

$$\therefore f^{-1} = f.$$

(ii)

$$\begin{aligned} f^{51}(4) &= f[f^{50}(4)] \\ &= f(4) \quad (\because f^2(x) = f^{-1}f(x) = x) \\ &= \frac{5-4}{1-4} = -\frac{1}{3} \end{aligned}$$

(iii)

$$\begin{aligned} g(x) &= 2x^2 + 4x + \lambda \\ &= 2(x+1)^2 - 2 + \lambda \geq -2 + \lambda \end{aligned}$$

For fg to be defined, $R_g \subseteq D_f = \mathbb{R} \setminus \{1\}$

$$\Rightarrow -2 + \lambda > 1 \Rightarrow \lambda > 3.$$

$$(-2, \infty) \xrightarrow{g} [-2 + \lambda, \infty) \xrightarrow{f} \left[\frac{7-\lambda}{3-\lambda}, 1 \right)$$

9(a) $S_n = S_{3n} - S_n \Rightarrow S_{3n} = 2S_n$

$$\frac{3n}{2} [2a - (3n-1)] = n [2a - (n-1)]$$

$$a = \frac{7n-1}{2}$$

9(b)

n th birthday	Amount of Mrs Lee's contribution (\$) on her daughter's n th birthday
0	1000
1	$1000(1.05)+1000$
2	$(1000(1.05)+1000)(1.05) + 1000 =$ $1000(1.05)^2 + 1000(1.05)+1000$
3	$(1000(1.05)^2 + 1000(1.05)+1000)(1.05)$ $+1000$ $= 1000[(1.05)^3 + (1.05)^2 + (1.05)+1]$
...	...
17	$1000[(1.05)^{17} + (1.05)^{16} + (1.05)^{15}$ $+ \dots + (1.05)^2 + (1.05)+1]$
18	$1.05 \times 1000[(1.05)^{17} + (1.05)^{16} + (1.05)^{15}$ $+ \dots + (1.05)^2 + (1.05)+1]$

Mrs. Lee: $a = 1000(1.05)$, $r = 1.05$

$$\text{Total amount} = \$ \frac{1050[(1.05)^{18} - 1]}{1.05 - 1}$$

$$= \$29500 \text{ (3 s.f.)}$$

n th birthday	Amount of Mr Lee's contribution (\$) on his daughter's n th birthday
1	$[1.05(500)+1.05(500)]$
2	$1.05(2 \times 1.05(500)) + 1.05^2(500)$ $= 3 \times 1.05^2(500)$
3	$1.05(3 \times 1.05^2(500)) + 1.05^3(500)$ $= 4 \times 1.05^3(500)$
...	...
17	$18 \times (1.05)^{17}(500)$
18	$\$1.05[18 \times (1.05)^{17}(500)]$

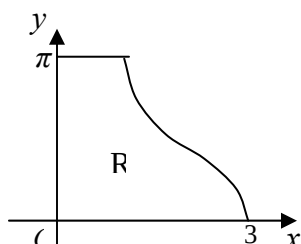
$$\text{Total amount} = \$[18 \times (1.05)^{18}(500)] = \$21700 \text{ (3 s.f.)}$$

$$10(i) \frac{d}{dx}(\sqrt{1-x^2}) = \frac{1}{2\sqrt{1-x^2}}(-2x) = -\frac{x}{\sqrt{1-x^2}}$$

$$\begin{aligned}\int \cos^{-1} x \, dx &= x \cos^{-1} x - \int x \left[-\frac{1}{\sqrt{1-x^2}} \right] \\ &= x \cos^{-1} x - \int \frac{1}{2} \frac{-2x}{\sqrt{1-x^2}} \, dx \\ &= x \cos^{-1} x - \sqrt{1-x^2} + C \quad (\end{aligned}$$

$$\therefore \int_{-1}^1 \cos^{-1} x \, dx = \left[x \cos^{-1} x - \sqrt{1-x^2} \right]_{-1}^1$$

$$(a) \, y = \cos^{-1}(x-2), \quad 1 \leq x \leq 3$$



Translate the graph of $y = \cos^{-1} x$ two units in the positive direction of x -axis to obtain the graph of $y = \cos^{-1}(x-2)$.

$$\text{Area of the region R} = \pi + \int_1^3 \cos^{-1}(x-2) \, dx$$

$$= \pi + \int_{-1}^1 \cos^{-1} t \, dt \quad \text{where } t = x-2 \Rightarrow dt = dx ;$$

$$x=1 \Rightarrow t=-1 \quad ; \quad x=3 \Rightarrow t=1$$

$$= 2\pi \quad (\text{Ans})$$

(b) Volume generated

$$= \int_0^\pi \pi x^2 \, dy = \pi \int_0^\pi (\cos y + 2)^2 \, dy$$

$$= \pi \int_0^\pi (\cos^2 y + 4 \cos y + 4) \, dy$$

$$= \pi \int_0^\pi \left[\frac{1}{2}(\cos 2y + 1) + 4 \cos y + 4 \right] \, dy$$

$$= \pi \left[\frac{1}{4} \sin 2y + 4 \sin y + \frac{9}{2} y \right]_0^\pi = \frac{9}{2} \pi^2 \quad (\text{Ans})$$

11(i) $d = -1$

$$y = -x + \frac{A}{x-1} = \frac{-x^2 + x + A}{x-1}$$

$$a = -1, \quad b = 1$$

(ii) When $x = 0$, $y = -c$

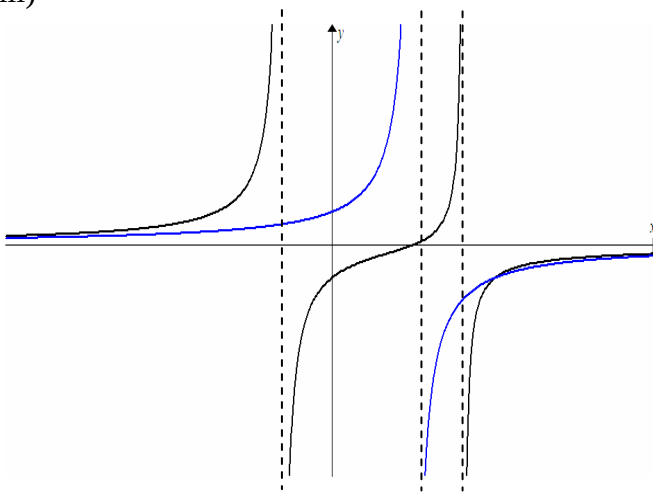
$$\text{When } y = 0, \quad \frac{-x^2 + x + c}{x-1} = 0$$

$$x = \frac{-1 \pm \sqrt{1+4c}}{-2}$$

$$= \frac{1 \pm \sqrt{1+4c}}{2}$$

Since $y < 0$ when $x = 0$, $-c < 0 \Rightarrow c > 0$

(iii)



Sketch graph of $y = \frac{1}{f(x)}$

$$\frac{1}{f(x)} = \frac{1}{1-x}$$

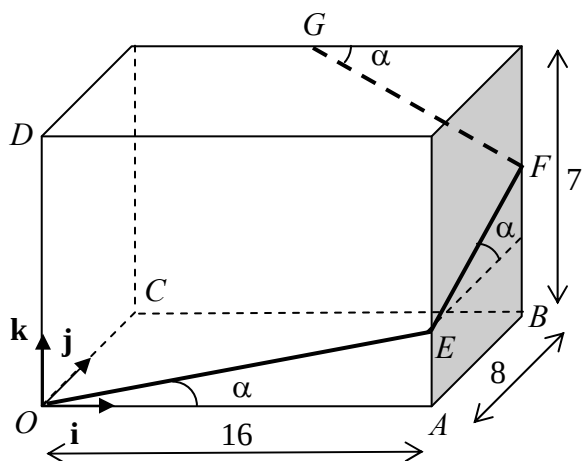
$$\frac{-x^2 + x + c}{x-1} = 1-x$$

$$-x^2 + x + c = -x^2 + 2x - 1$$

$$x = c+1$$

$$\frac{1}{2}(1 - \sqrt{1+4c}) < x < 1 \quad \text{or} \quad \frac{1}{2}(1 + \sqrt{1+4c}) < x < c+1$$

12



$$(i) \quad \vec{OE} = \begin{pmatrix} 16 \\ 0 \\ 4 \end{pmatrix}, \vec{OF} = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}, \vec{OG} = \begin{pmatrix} 12 \\ 8 \\ 7 \end{pmatrix}$$

(ii)

$$\text{Area of } \triangle OEG = \frac{1}{2} |\vec{OE} \times \vec{OG}| = \frac{1}{2} \left| \begin{pmatrix} 16 \\ 0 \\ 4 \end{pmatrix} \times \begin{pmatrix} 12 \\ 8 \\ 7 \end{pmatrix} \right| = 16\sqrt{21}$$

Area of $\triangle OEG = \frac{1}{2} (OE)(d) = 16\sqrt{21}$, where d is the perpendicular distance from G to OE
 $d = 8.89$

(iii) Equation of the plane containing points E , F and G :

$$\vec{EF} = \begin{pmatrix} 0 \\ 8 \\ 2 \end{pmatrix}; \vec{EG} = \begin{pmatrix} -4 \\ 8 \\ 3 \end{pmatrix}$$

$$\mathbf{n} = \begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix} \times \begin{pmatrix} -4 \\ 8 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \\ 16 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}$$

$$\therefore \text{ equation of the plane is } \mathbf{r} \cdot \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 16 \\ 0 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} = 32$$

(iv) From (iii), equation of plane containing the points E, F

and G is $\mathbf{r} \cdot \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} = 32$.

Since $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ does not satisfy the equation of plane EFG ,

the points O, E, F and G do not lie on the same plane.

(v) The required plane Π will be parallel to OF and the line of intersection of planes OEG and EFG .

i.e. the plane Π is parallel to OF and EG .

Thus, the equation is

$$\mathbf{r} = \lambda \begin{pmatrix} 8 \\ 4 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ 8 \\ 3 \end{pmatrix},$$

where $\lambda, \mu \in \mathbb{R}$.